ECON330 Exam 2 — AI Solver Packet

Built from your Exam 2 outline and the practice exam key (Chapters 8, 9, 12, 13).
Everything below the divider is meant to be pasted into an AI model.

How to use this

- Paste **Part 1** (the instruction prompt) first, then **Parts 2–7** as the knowledge base, all in one message.
- Send each exam question separately. Screenshot any graph or table, since the AI can't guess missing data.
- For calculations, spot-check the AI against the formulas in Part 7.
- This packet has no actual exam questions. It has your notes, formulas, traps, and the practice-exam solutions as worked patterns.

Part 1 — Instruction prompt (paste first)

You are my solver for ECON330 Exam 2 (Mishkin: Ch 8 financial structure, Ch 9 banking, Ch 12 financial crises, Ch 13 the Fed). Treat the course notes below as the source of truth for definitions, formulas and conventions. If a question conflicts with them, tell me.

Rules:

- Multiple choice: give the letter, then 1–2 lines why, then one line on why the most tempting wrong answer is wrong.
- Calculations: write the formula, plug in numbers, give units ($ millions, %, percentage points). Convert rates to decimals (3 points = 0.03). Re-check arithmetic before answering.
- T-accounts: draw Assets | Liabilities with signed changes. Always check that assets and liabilities change by the same amount.
- Graphs: name the axes, curves, shifts and labeled points (D1, D2, S, P1, P2, P3).
- Written questions: answer each lettered part separately, show work, and use the key terms from the notes.
- Classification questions (which problem / job / stage / tool): name the category first, then justify in one line.
- If data is missing (a table, graph or number), ask me before guessing.

- Watch the common traps listed in Part 7.

Part 2 — Chapter 8: Economic Analysis of Financial Structure

Facts and data

- Direct finance = firm sells stocks and bonds directly to savers. Indirect finance = savers deposit in a bank, and the bank lends to the firm.
- U.S. external funds (1970–2000): bank loans 18% + nonbank loans 38% = **56% loans (indirect)**. Bonds 32% + stocks 11% = **43% securities**. Stocks are the smallest source. The largest single bar is **nonbank loans (38%)**, not bank loans. Germany and Japan: bank loans above 75%.

The 8 facts

1. Stocks are not the most important source of external funds.
2. Marketable securities (stocks and bonds) are not the main way firms finance.
3. Indirect finance is far more important than direct finance.
4. Banks are the most important source of funds in many countries.
5. The financial system is heavily regulated.
6. Only large, well-known firms have easy access to securities markets.
7. Collateral is common in debt contracts.
8. Debt contracts are complicated, with many restrictive covenants.

Two causes

- **Transaction costs** = time and money to carry out a deal (find a borrower, write a contract, brokerage fees). Small savers face a high cost per dollar and can't diversify. Intermediaries fix this with economies of scale, expertise and liquidity services (checking accounts). Explains part of Facts 3 and 4.
- **Asymmetric information** = one side knows more.
  - **Adverse selection** = BEFORE the deal, hidden type. The riskiest borrowers are the most eager to borrow.
  - **Moral hazard** = AFTER the deal, hidden action. The borrower takes risks the lender wouldn't approve, since the lender bears part of the loss (car loan spent at a casino).

    - Agency theory = study of how information problems affect behavior.

Adverse selection: the lemons problem

- Used cars: half good ($10,000), half lemons ($6,000), and only the seller knows which. Buyers pay the average, ½(10,000) + ½(6,000) = $8,000. Good sellers won't sell a $10,000 car for $8,000 and leave. Buyers realize mostly lemons are left, offer less, and the market shrinks.
- **Rule: if good sellers' minimum exceeds the average, the price falls to the LEMON value.**
    - Example: good cars $12k, lemons $6k, good owners won't sell below $11k. Average = $9k, which is below $11k, so good owners leave. Only lemons remain, so **price = $6,000**. ($9,000 is only the first offer.)
- Securities: investors can't tell good firms from bad, so they pay the average price. Good firms' securities are undervalued, good firms don't issue, and securities markets stay small (explains Facts 1 and 2).

Four ways to reduce adverse selection

1. **Private production and sale of information** (Standard & Poor's, Moody's). Problem: **free riders** copy buyers' trades or read the report. Producers can't earn enough, so too little information is made. Explains Facts 1 and 2.
2. **Government regulation** (SEC). Firms must disclose sales, assets and earnings. Standard accounting principles plus audits, and fraud is punished. Limit: Enron (2001) hid debt with complex accounting. Regulation reduces but can't eliminate the problem. Explains Fact 5.
3. **Financial intermediaries (banks)**, the "car dealers of finance": experts at screening good from bad borrowers. Bank loans are private and not traded, so others can't copy the information and the bank keeps the profit. This solves both screening and free riding, so banks dominate. Explains Facts 3, 4 and 6. Large, well-known firms: investors already have information, so there is little lemons problem and they can sell securities (Fact 6).
4. **Collateral and net worth**. Collateral = property pledged if the borrower defaults (house for a mortgage, car for an auto loan). The lender sells it and loses less. Net worth = what the firm owns minus what it owes, and high net worth works like collateral. Explains Fact 7. Result: it seems "only people who don't need money can borrow."

Moral hazard in equity (principal–agent problem)

- **Principals** = shareholders (own the firm, want maximum profit). **Agents** = managers (run the firm, may own a small share). Ownership and control are separated, so managers pursue their own benefit.
- Example: a manager owns 10%, and a makeover costs the firm $10,000. She pays 10% × $10,000 = $1,000 but gets the full benefit, so she overspends. **A smaller stake makes the problem worse.**
- Fixes:
  - **Monitoring** = costly state verification (checking what managers do). Other shareholders free-ride, so equity is less attractive (Fact 1).
  - **Regulation**: accounting and disclosure rules make it harder to hide profits (Fact 5).
  - **Venture capital**: takes board seats to monitor; shares cannot be sold, so no one free-rides (Fact 3). VCs still pay to verify.
  - **Debt contracts**: fixed payments; the lender checks only if the borrower fails to pay, so less verification is needed (Fact 1). Debt vs equity: debt needs less verification.

Moral hazard in debt

- The borrower keeps the upside and the lender bears the downside, so the borrower prefers a risky project.
- Example: borrow $1,000, repay $1,100.
  - Safe project pays $1,200 for sure: borrower keeps $100, lender gets $1,100.
  - Risky project pays $2,400 or $0 (50% each): borrower expects 0.5 × (2,400 − 1,100) = $650, lender expects 0.5 × 1,100 = $550.
  - Borrower prefers risky ($650 > $100), and the lender is worse off ($550 < $1,100).
- Fixes:
  - **Net worth and collateral**: more own money at stake means more to lose.
  - **Restrictive covenants** (Fact 8), 4 types: discourage bad behavior (limit new borrowing); encourage good behavior (keep minimum net worth); keep collateral valuable (maintain insurance, e.g. a mortgage requiring home insurance); provide information (regular audited reports).
  - **Monitoring by banks**: covenants must be monitored and enforced, which is costly. Bondholders free-ride on each other; a bank with a private loan does not.

Summary: problem → tool → facts

- Adverse selection: private info (1, 2), regulation (5), intermediation (3, 4, 6), collateral/net worth (7).
- Moral hazard in equity: monitoring (1), regulation (5), VC (3), debt contracts (1).
- Moral hazard in debt: collateral/net worth (6, 7), covenants (8), bank monitoring (3, 4).
- Fact 1: lemons + free riding + costly verification. Fact 2: lemons. Fact 3: lower transaction costs + screen/monitor. Fact 4: private loans, no free riders. Fact 5: disclosure. Fact 6: well known. Fact 7: collateral cuts losses. Fact 8: covenants limit moral hazard.

Application: insurance

- House $800,000. Flood chance 1 in 40 without a seawall, 1 in 100 with one. **Fair premium = probability × loss**: 800,000 ÷ 40 = $20,000; 800,000 ÷ 100 = $8,000. If the insurer can see the seawall, the premium falls $12,000, so the owner builds it. If prevention is hidden, full coverage weakens care (moral hazard).

Application: development and growth

- Weak property rights mean assets can't be used as collateral ("dead capital," Hernando de Soto). Slow courts make covenants unenforceable. Weak accounting means little reliable information (lemons). State-owned banks and directed credit send funds to favored projects (taxpayers cover losses).
- Result: adverse selection and moral hazard stay high, so there is less lending, capital goes to less productive uses, and growth is slow.

China

- Growth framework: $Y/L = A \times f(K/L, H/L, N/L)$ (physical capital, human capital, resources, technology).
- China 1978–2010: ~10% growth per year, with credit from state banks and weak property rights and disclosure.
- Why it grew anyway: a saving rate of ~40% of income (heavy investment) and workers moving from farms to factories.
- Why it may not last: state banks have weak incentives to lend to the most productive firms. As China gets richer, growth depends on allocating capital well, which needs a better financial system.

- China is NOT a counterexample to "financial development drives growth." Rapid growth does not mean the financial system was efficient.

Quick classification (Ch 8)

- Lender can't tell risky applicants = adverse selection (screening, collateral).
- Firm secretly switches to a riskier project = moral hazard (covenants, net worth, monitoring).
- CEO with a 2% stake buys a jet = moral hazard in equity (monitoring, VC, debt).
- Investors copy a research firm's picks = free rider.
- Mortgage requires home insurance = restrictive covenant.

Part 3 — Chapter 9: Banking and Management of Financial Institutions

Balance sheet

- **Assets = Liabilities + Capital**. Capital = Assets − Liabilities (the cushion).
- Example: reserves 20 + securities 10 + loans 80 = assets 110. Deposits 100 + capital 10 = 110. Loans lose 6, so assets = 104. Deposits are still owed in full (100), so capital = 104 − 100 = 4.
- Deposits are the bank's liability (and your asset). Liabilities (sources of funds): deposits ~77%; borrowings (discount loans from the Fed, federal funds from other banks); bank capital (owners' money, not a promise to repay).
- Assets, most liquid to least liquid: reserves (lowest return) → securities → loans (52% of assets; highest return; main source of profit).
- Three goals: profit, liquidity, safety.

T-accounts

- Deposit $100 cash: reserves +100, checkable deposits +100.
- Deposit a $100 check from another bank: your bank reserves +100 / deposits +100; the other bank reserves −100 / deposits −100.
- **Rule: gain deposits → gain equal reserves; lose deposits → lose equal reserves.**

How a bank profits

- **Asset transformation**: borrow short, lend long (deposits can leave any day, loans last years), which creates liquidity risk and interest-rate risk.
- New deposit D, reserve ratio r: required reserves +rD, excess reserves +(1−r)D, then loans +(1−r)D, deposits +D. **Assets rise by the full D.**
  - Example: new deposit $1,000,000, reserve ratio 20%. Reserves +$200,000, loans +$800,000, deposits +$1,000,000, assets +$1,000,000.
- Spread example: $1,000,000 of deposits at 2% = $20,000 cost. Lend $900,000 at 6% = $54,000 income. Net interest income = $34,000.

Six management jobs

1 Liquidity • 2 Assets • 3 Liabilities • 4 Capital • 5 Credit risk • 6 Interest-rate risk

Job 1: Liquidity (deposit outflow)

- With a buffer: reserves 20, deposits 100; a $10m outflow leaves reserves 10, deposits 90; required = 10% × 90 = 9; excess 1. No changes needed. Excess reserves are insurance.
- No buffer: reserves 10, deposits 100; a $10m outflow leaves reserves 0, deposits 90, required 9, so **shortfall 9**.
- **Steps for an outflow X:**
  1. Required = r × D; excess = R − required.
  2. T-account: Reserves −X | Deposits −X.
  3. New required = r × (D − X).
  4. **Shortfall = new required − (R − X)**. Use the NEW deposit level.
  - Excess reserves fall by X(1 − r), not X.
  - Largest outflow with no changes needed: X = (R − rD) ÷ (1 − r).
- Worked example (practice Written Q1): reserves 26, deposits 184, r = 10%.
  - Required = 18.4; excess = 26 − 18.4 = **7.6**.
  - $18 outflow: reserves 26 − 18 = 8; deposits 184 − 18 = 166. New required = 16.6. **Shortfall = 16.6 − 8 = $8.6**.
  - Why not 18 − 7.6: required reserves also fall by 1.8, so excess falls by 18 − 1.8 = 16.2, from +7.6 to −8.6.
- Practice MC example: reserves $50m, deposits $200m, outflow $50m, r = 10%. Reserves = 0, deposits = 150, required = 15, **shortfall $15m**.

- **4 ways to get reserves (cheapest → costliest):**
  1. Borrow from other banks: reserves +, borrowings +; cost = fed funds rate.
  2. Sell securities: reserves +, securities −; cost = brokerage fees. T-account: Reserves +s, Securities −s, liabilities unchanged (asset side only).
  3. Borrow from the Fed: reserves +, discount loans +; cost = discount rate.
  4. Reduce or call in loans: reserves +, loans −; cost = angry customers, and loans sell at a discount (buyers fear adverse selection). **Worst option.**
  - A bank can NOT "reduce deposits" to fix a shortfall.
- Excess reserves: benefit = meet withdrawals without borrowing or selling. Cost = excess × (loan rate − rate on reserves). Example: excess $10m, loan rate 5%, reserve rate 4%, so 10m × (5% − 4%) = $100,000 per year.

Job 2: Assets

- Earn high returns (loans to borrowers who pay high rates but are unlikely to default; good securities). Keep risk low (diversify across borrowers, industries, places). Stay liquid (reserves and liquid securities, with lower return). The bank can't chase the highest return while ignoring default risk and liquidity.

Job 3: Liabilities

- Then: passive (deposits fixed, adjust assets). Now: active, so target asset growth and raise funds as needed (negotiable CDs, fed funds).
- Example: a creditworthy firm wants a loan but the bank has no excess reserves → issue negotiable CDs or borrow fed funds.
- Trade-off: market funding costs more and can dry up in a crisis.

Job 4: Capital

- More capital = loss-absorbing cushion and a lower chance of failure.
- Example: both banks lose $5m. The bank with capital 10 → capital 5 (survives). The bank with capital 4 → capital −1 (insolvent, assets < liabilities).
- **ROA = profit ÷ assets; ROE = profit ÷ capital; EM (equity multiplier) = assets ÷ capital; ROE = ROA × EM.**
  - Assets $100m, profit $1m (ROA = 1%). Capital $10m: EM = 10, ROE = 10%. Capital $5m: EM = 20, ROE = 20% (but a loss just over $5m wipes it out).

  - ROA 1.2%, capital = 8% of assets: EM = 1 ÷ 0.08 = 12.5; **ROE = 1.2% × 12.5 = 15%.** The multiplier is 1 ÷ capital ratio, not 8.
  - Practice Written Q1(d): ROA = 2/200 = 1%; ROE = 2/16 = 12.5%; EM = 200/16 = 12.5 (check: 1% × 12.5 = 12.5%).
- Capital buys safety at the cost of lower ROE. **Higher capital lowers ROE for a given ROA.** Banks hold too little capital without regulation: higher ROE, plus the safety net and deposit insurance shift failure costs off owners. Hence capital requirements.
- Raise capital: issue stock, cut dividends, shrink assets. Lower capital: buy back stock, raise dividends, grow assets with borrowed funds.
- Credit crunch (2007–09): MBS losses → capital falls → hard to raise → banks cut lending.

Job 5: Credit risk (Ch 8 tools)

- Before the loan: screening and specialization. After the loan: monitoring and covenants. Long-term relationships and loan commitments make screening and monitoring cheaper.
- **Collateral and compensating balances** (a required checking balance): they reduce losses on default, and the bank watches the firm's checking activity (monitoring, which cuts moral hazard). Not a high-return tool.
- **Credit rationing** (a firm offers 15% when others pay 6%): the high rate attracts risky borrowers (adverse selection) and pushes riskier projects (moral hazard), so the bank refuses or lends less rather than raising the rate.

Job 6: Interest-rate risk

- **Gap analysis (profits): gap = RSA − RSL (rate-sensitive assets minus rate-sensitive liabilities); Δprofit = gap × Δi** (decimal: 3 points = 0.03).
  - Positive gap: rates up → profit up. Negative gap: rates up → profit down.
  - RSA $20m, RSL $50m: gap = −30. Rates +5 points: −30 × 0.05 = **−$1.5m**. Rates −5 points: −30 × (−0.05) = **+$1.5m**.
  - RSA $60m, RSL $40m: gap = +20. Rates −3 points: 20 × (−0.03) = **−$0.6m** (profit falls). A positive gap means profits fall when rates fall.
- **Duration analysis (bank value): ΔV ≈ −duration × Δi × value; ΔNW = ΔA − ΔL.**
  - Assets $100m (duration 3), liabilities $90m (duration 2), rates +5 points: ΔA = −3 × 0.05 × 100 = −15; ΔL = −2 × 0.05 × 90 = −9; **ΔNW = −15 − (−9) = −$6m**.

    - Assets $200m (duration 2), liabilities $180m (duration 4), rates +2 points: ΔA = −2 × 0.02 × 200 = −8; ΔL = −4 × 0.02 × 180 = −14.4; **ΔNW = −8 − (−14.4) = +$6.4m** (liabilities lose more).
    - The longer-duration side loses more when rates rise.
- Gap = profits; duration = bank value. Gap and duration measure interest-rate risk, NOT trading risk.
- Reduce risk: shorten asset duration (variable-rate or short-term assets), lengthen liability duration (long CDs), hedge with futures, options, swaps. Expect rates up: short-term/variable-rate assets, lengthen liabilities. Expect rates down: longer-term fixed-rate assets (lock in).

Off-balance-sheet activities

- Loan sales (keep profit plus servicing fees). Fee income: FX trades, guarantees, backup lines of credit (= loan commitments), overdraft privileges, standby letters of credit. Trading: futures, options, swaps, FX. Not risk-free (guarantees add risk).
- **Rogue traders (Barings 1995)**: principal–agent problem. The trader gets a bonus if bets win and the bank bears the loss; the same person trades and keeps the records, so losses get hidden.
- Controls: separate trading from bookkeeping; limit exposure; **value-at-risk (VaR)** = largest likely loss over a period at a given probability; **stress tests** = losses if an unusual combination of bad events occurs. Neither VaR nor stress tests guarantee safety.

Which job?

- Unexpected $10m withdrawal = liquidity.
- Stops lending to one industry = assets (diversify).
- Issues negotiable CDs for a loan = liabilities.
- Regulators say the capital ratio is too low = capital.
- Borrowers miss payments = credit risk.
- Short-term funding cost rises faster than loan income = interest-rate risk (negative gap).

Part 4 — Chapter 12: Financial Crises

What a crisis is

- Financial frictions = adverse selection + moral hazard. When they rise, credit is scarce and expensive.
- **Financial crisis = a sharp jump in frictions**: asset prices fall, many firms fail, and markets stop moving funds to productive uses. A stock decline alone, with normal lending, is not a crisis. Falling asset prices can trigger one, but the crisis is the breakdown of lending.
- Chain: asymmetric information worsens → frictions rise → lenders cut credit → spending falls → the economy contracts.
- Two things keep frictions low: (1) borrowers' net worth and collateral (less moral hazard; lenders recover losses); (2) healthy banks (they screen and monitor; bank capital absorbs losses). A crisis starts when net worth falls, banks weaken, or uncertainty rises.

Three stages

- **Stage 1: Initiation.** Something lowers net worth or raises uncertainty, so lending falls. Three ways:
  - **1a Credit boom and bust**: innovation or liberalization → lending boom with weak risk management → loan losses → bank capital falls → deleveraging (cutting lending). The safety net means little market discipline. Loan losses come out of capital, so the bank must shrink assets.
  - **1b Asset-price boom and bust**: prices fall, debt doesn't → net worth destroyed → moral hazard up.
  - **1c Jump in uncertainty**: stock crash, big firm fails, recession starts → screening is harder → adverse selection up, credit spread widens.
- **Stage 2: Banking crisis.** Losses push banks toward insolvency; depositors can't tell good banks from bad, so runs and panic follow. Fire sales push prices down. Fewer banks, less information, less lending. Recovery is slow.
- **Stage 3: Debt deflation.** An unanticipated fall in prices raises the real debt burden and destroys net worth a second time, so the recession lasts longer. Not every crisis has it. If the fall were expected, lenders would have charged lower nominal rates.

- **Credit spread = Baa rate − Treasury rate**, the "crisis thermometer." Example: Baa 6.5%, Treasury 4.0% → spread 2.5 points. Baa jumps to 9.0% → spread 5.0 points (**rose 2.5**). Give the INCREASE (peak − start), not the peak level. Practice Q21: peak 5.6, start 2.0 → increase 3.6 points. A wider spread = worse information problems.
- **TED spread** = 3-month interbank rate − 3-month T-bill: ~40 bp pre-crisis, ~240 in Aug 2007, 450+ after Lehman.

Formulas (Ch 12)

- **Home equity = value − mortgage; % change = Δequity ÷ original equity.**
  - House $300k, mortgage $240k → equity $60k. Price −10%: house $270k → equity $30k → change = −30 ÷ 60 = **−50%**. Price −20%: house $240k → equity $0 → **−100%**.
  - Peak example: bought $400k, mortgage $360k. Price +10% → $440k (the peak). Then a 25% fall from the peak: 0.75 × 440 = $330k. Equity = 330 − 360 = **−$30k** (underwater). **The bust % applies to the PEAK price, not the original.**
  - Practice Written Q2(b): bought $300k, mortgage $270k. +20% → $360k (peak), equity $90k. Bust of 30% of the peak: 0.70 × 360 = $252k, equity = 252 − 270 = **−$18k**.
- **Bank capital lost = loss ÷ old capital (not assets).** Assets $100m, capital $8m, loss $6m → new capital $2m; % of capital lost = 6 ÷ 8 = **75%** (but only 6% of assets). Practice Q20: loss $15m, capital $24m → 15 ÷ 24 = **62.5%** (loss is only 3.75% of $400m assets).
- **Debt deflation: real debt = nominal debt ÷ new price level.** Owes $100,000; price level falls 10% (new level 0.90) → 100,000 ÷ 0.90 = **$111,111** (+11.1%).
- **Repo haircut h: loan = (1 − h) × collateral → collateral = loan ÷ (1 − h); max assets = capital ÷ h.**
  - Borrow $170m, haircut 15% → collateral = 170 ÷ 0.85 = **$200m** (not 170 × 1.15 = 195.5).
  - Capital $5m: haircut 5% → max assets = 5 ÷ 0.05 = $100m. Haircut 20% → 5 ÷ 0.20 = $25m. Must sell 100 − 25 = **$75m**. Don't add the haircut to the loan.
- **Currency mismatch: peso debt = dollar debt × exchange rate.** Acts like debt deflation. Firm owes $5m, assets 55m pesos. Rate moves from 8 to 12 pesos per $. Before: debt = 5 × 8 = 40m, net worth = 55 − 40 = 15m. After: debt = 5 × 12 = 60m, net worth = 55 − 60 = **−5m pesos**. The dollar debt didn't change, but its peso value rose 50%.

- **CDO tranches: losses hit equity first → mezzanine → senior.** Subtract the equity tranche first, then divide the remainder by the mezzanine size.
  - Pool loses $50m. Equity $20m, mezzanine $40m: equity absorbs 20 (wiped out). Remaining 30 goes to mezzanine: 30 ÷ 40 = **75%**; senior loses 0. (25% would be the pool-wide loss rate. Wrong.)
  - $100m pool: equity $5m, mezzanine $15m, senior $80m. Loss $3m → equity loses 3 ÷ 5 = 60%. Loss $10m → equity 100%, the other 5 hits mezzanine = 5 ÷ 15 = 33%. Loss $25m → equity 100%, mezzanine 100%, the other 5 hits senior = 5 ÷ 80 = 6.25%.

Great Depression

- Stage 1: stocks doubled 1928–29, Fed raised rates; crash Oct 1929; by 1932 stocks were down ~90%.
- Stage 2: bank panics late 1930–33; ~1/3 of banks failed.
- Stage 3: prices −25% (1930–33); unemployment ~25%.

2007–09

- Subprime mortgages (weak credit, little or no down payment); securitization (MBS → CDOs sold to investors); housing bubble. Credit boom (1a) + housing bubble (1b).
- **Originate-to-distribute (O2D)**: a broker makes the loan (paid per loan) → sold to an investment bank → bundled into MBS/CDOs → rated → investors hold the risk. Brokers are paid for volume, not quality, so screening is weak. Borrowers have little down payment and little to lose. Rating agencies are paid by issuers and helped design the securities (conflict of interest). Whoever screens doesn't bear the loss = moral hazard / principal–agent.
- Prices +60% (2002–06), −1/3 (2006–09); negative equity → defaults; MBS/CDOs lose value; capital shrinks.
- **Run on shadow banks (repo)**: doubts → lenders raise haircuts → fire sales → prices fall → repeat. Lenders raising haircuts = a run.
- Timeline 2008: March Bear Stearns sold to JPMorgan; summer Fannie/Freddie rescue (taken over in Sept); mid-Sept Lehman fails, Merrill sold, AIG rescued, a money fund "breaks the buck"; late Sept WaMu fails, Congress votes TARP down; Oct 3 TARP ($700bn) passes.
- **No Stage 3**: policy kept prices from falling. Peak unemployment ~10% (vs 25%).

- Policy: lender of last resort (Fed); TARP (bank stakes); deposit insurance raised to $250,000 plus money funds guaranteed; fiscal stimulus $787bn (Feb 2009). Trade-off: rescues limit damage now but encourage risk later.
- Debate: **Taylor** says the Fed held rates far below the Taylor rule in 2003–06, giving cheap credit, so the Fed should lean against bubbles. **Bernanke** says a global savings glut, risky mortgage products and weak regulation caused it, so use regulation.

Housing graph (practice Written Q2)

- Axes: home price (y), homes sold per month (x). Draw S and D1, with equilibrium P1.
- Looser lending standards: D shifts RIGHT (D1 → D2), price rises P1 → P2.
- Bust: D shifts LEFT, price falls to P3 < P2. Label curves, shifts and prices. (A rightward supply shift from foreclosure sales is also acceptable.)
- Why lenders loosened (O2D): they earn fees, sell the loan, and bear little default risk, so they have a weak incentive to screen (principal–agent).

More crises

- Europe sovereign debt: fear that governments (Greece first) can't repay, so bond rates soar. Banks held government bonds, so capital fell. Austerity and deep recessions.
- COVID (2020): sudden jump in uncertainty (1c); spreads spiked and markets froze. Short because of fast Fed lending and fiscal relief.
- Emerging markets: Stage 1 credit boom (path A) or fiscal imbalance (path B) → Stage 2 currency crisis → Stage 3 full-fledged crisis. Debt is in dollars, revenue in pesos. Mexico 1994–95 (A; peso lost ~half); South Korea 1997–98 (A; chaebols, short-term foreign funds); Argentina 2001–02 (B; banks forced to hold government bonds). Prevention: strong supervision, disclosure, limit currency mismatch, liberalize gradually.

Prevention

- **Microprudential**: is ONE institution safe? Capital requirements, supervision, exams. Blind spot: safe for one bank, dangerous when all do it (fire sales lower prices, capital and collateral for everyone).
- **Macroprudential**: is the WHOLE system safe? Capital buffers that rise in booms, loan-to-value limits, system-wide stress tests; lean against credit booms.

- **Dodd-Frank 2010**: CFPB (consumer protection); resolution authority (seize or wind down a giant firm); FSOC (systemic risk); Volcker Rule (limits banks' own-money trading); standardized derivatives via clearinghouses; stress tests for large banks.
- **Too big to fail**: creditors expect a rescue, stop watching, and the firm takes more risk (moral hazard). Fixes: break up (lose scale), higher capital (less lending, lower ROE), Dodd-Frank tools (will they be used?). Still open: executive pay, Fannie/Freddie, rating-agency conflicts, overregulation.

Great Depression vs 2007–09

- Stage 1: stock crash / mortgage credit boom + housing bust. Stage 2: bank panics, 1/3 of banks fail / run on shadow banks, big firms fail. Stage 3: yes (prices −25%) / no. Peak unemployment: ~25% / ~10%.

Which stage?

- House prices −30%, equity wiped = 1b.
- Repo haircuts up, shadow banks sell = Stage 2.
- Price level falls 10% more than expected = Stage 3.
- Huge firm suddenly fails = 1c.
- Banks lend heavily after rules are relaxed, loans sour = 1a.
- Peso falls, dollar debts soar = emerging-market Stage 3.

Part 5 — Chapter 13: Central Banks and the Federal Reserve

History

- 1791–1811 First Bank of the U.S. (charter not renewed). 1816–1836 Second Bank of the U.S. (not renewed). 1836–1913 no central bank, frequent panics. 1907 severe panic → need for a lender of last resort. **1913 Federal Reserve Act.**
- Needed a lender of last resort but feared a powerful central bank. Checks and balances: Washington vs 12 regional banks; public (presidential appointees) vs private (banks own stock); bankers vs everyone else. Power split on paper, moved to Washington in practice.

Structure

- **Board of Governors (7, D.C.)**: appointed by the President, confirmed by the Senate. **14-year terms, not renewable; one term expires every 2 years. Chair: 4-year term, renewable.** Votes on the FOMC (all 7); sets reserve requirements and interest on reserve balances; effectively sets the discount rate (approves Reserve Banks' proposals); sets margin requirements; approves Reserve Bank presidents and budgets; appoints 3 directors; regulates and supervises banks. Long staggered terms protect from political pressure.
- **12 Federal Reserve Banks, 9 directors each:**
  - Class A (3): professional bankers, elected by member banks.
  - Class B (3): industry, labor, agriculture, consumers; elected by member banks.
  - Class C (3): public; appointed by the Board of Governors; the board chair comes from C.
  - **President: chosen by Class B and C directors, approved by the Board of Governors.** Bankers (A) don't vote. Presidents are not political appointees.
- Member banks: own Reserve Bank stock, elect 6 directors; dividends limited by law; run for the public. National banks must be members; state banks may join; ~1/3 of commercial banks. DIDMCA 1980: only members held costly interest-free reserves, so banks left; now the same reserve rules apply to all depository institutions. Today requirements are 0% and reserves earn interest.
- Reserve Banks: clear checks, manage currency, make discount loans to district banks, supervise (holding companies, state member banks), review mergers, do data and research.
- Districts: 1 Boston, 2 NY, 3 Philadelphia, 4 Cleveland, 5 Richmond, 6 Atlanta, 7 Chicago, 8 St. Louis, 9 Minneapolis, 10 Kansas City, 11 Dallas, 12 San Francisco. **Maryland = District 5, Richmond (Baltimore branch).**
- NY Fed is special: supervises many of the biggest institutions; has the open market desk (trades for the FOMC); does FX intervention (for the Fed and Treasury); sits next to Wall Street; permanent FOMC vote (vice chair); only Reserve Bank in the BIS; holds gold.

FOMC and who controls the tools

- **12 voters = 7 governors (every meeting) + NY president (every year) + 4 other presidents (rotate).** 19 at the table (7 + all 12 presidents), 12 votes. Meets 8 times a year (~every 6 weeks); decides the fed funds rate target, open market operations and

the balance sheet.

- Meeting flow: staff report on markets → staff forecast → go-around 1 (economy) → go-around 2 (policy) → Chair proposes, 12 vote → statement and press conference.
- **Beige Book** = business stories from all 12 districts, before each meeting. **Tealbook** = confidential staff forecast and policy alternatives (replaced the Green/Blue Books in 2010).
- Chair: 1 vote of 12, **no veto**; in practice the most powerful (spokesperson who testifies to Congress, sets the agenda, controls staff, builds consensus).
- Tools: open market operations → FOMC (NY desk trades). Discount rate → Reserve Banks propose, Board approves (effectively decides). Reserve requirements → Board (0% since Mar 2020). Interest on reserves → Board (keeps fed funds in the target range).

Independence

- **Instrument independence** = freedom to choose how to use the tools (Fed: strong). **Goal independence** = freedom to choose the goals (Fed: partial; Congress sets the dual mandate and can change it by law).
- Protects independence: 14-year nonrenewable terms; own income from securities (no budget from Congress).
- Limits: Congress can change structure and goals by law; the President appoints governors and the Chair.
- **Annual congressional appropriations = LESS independent.** Testifying to Congress = accountability, not loss of independence.
- Case FOR independence: politicians' short horizons; **political business cycle** (expand before an election, tighten after → inflation bias); an independent Fed refuses to print money for deficits; principal–agent (politicians' incentives to ignore the long run are stronger).
- Case AGAINST: undemocratic; less accountable (voters can't remove); harder monetary/fiscal coordination; imperfect record (Depression mistakes, 1970s inflation).
- **Bureaucratic behavior theory**: a bureaucracy maximizes its own power and prestige, so the Fed fights to keep its authority, is secretive, avoids conflict with Congress and the President, and is slow to raise rates if that would provoke conflict. Giving away budget control or authority is the opposite of what the theory predicts.

- Evidence: more independent central banks mean lower inflation on average, with no worse unemployment or output. Worldwide trend: independence plus transparency. Most economists favor instrument independence with accountability. Political pressure is real (public attacks on rate hikes 2018–19); counter-example: the 1990s Treasury no-comment policy.

Transparency

- Before 1994: decisions not announced. 1994: decisions announced after each meeting. Statements with explanation and forward guidance; minutes 3 weeks after. 2012: explicit 2% inflation goal; dot plot. Today: press conference every meeting. **"Transparency is the price of independence."**

Other central banks

- **ECB**: Executive Board (president, VP, 4 members; 8-year terms) = Board of Governors. **Governing Council** (Executive Board + national central bank governors; meets every 6 weeks) = FOMC. National central banks = Reserve Banks (control budgets). Goal: price stability, in the Maastricht Treaty; changing it needs every member country. **Most independent central bank in the world (instrument + goal).** The Fed has a dual mandate (max employment + stable prices) set by Congress (changeable by law).
- **Bank of Canada**: the government has final responsibility; a directive is possible but never used; the inflation target is set jointly → **less instrument and goal independence than the Fed**.
- **Bank of England**: sets rates since 1997; the government sets the target = instrument, not goal independence.
- **Bank of Japan**: instrument independence since 1998; the government can ask for a delay but has no veto.
- Trend: worldwide toward instrument independence. Don't confuse Bank of Canada with the ECB.

True / false traps

- Chair can veto the FOMC? **False.**
- All 12 presidents vote every meeting? **False** (5 vote: NY + 4 rotating).
- Fed depends on Congress for its budget? **False.**
- Member banks control the Fed because they own stock? **False.**
- ECB goal changed by ordinary law? **False** (treaty).

- Maryland banks are in the Richmond district? **True.**

Fed by the numbers

1913 Act • 12 Reserve Banks • 7 Board members • 14-year governor term • 4-year Chair term • 12 FOMC voters (7 + 5) • 8 meetings a year • 9 directors per Reserve Bank • 0% reserve requirement since Mar 2020 • 2% goal since 2012

Part 6 — Practice-exam answer patterns (30 MC + 2 written)

These are the question types in the practice exam, with the reasoning the instructor expects. Expect the real exam to follow the same patterns with different numbers.

Chapter 8

1. Indirect finance is more important than direct: loans 18% + 38% = 56% vs securities 32% + 11% = 43%. The largest single bar is nonbank loans, not bank loans. (D)
2. Lemons: good $12k / lemon $6k, good owners need $11k. Average $9k is too low, so only lemons remain. **$6,000.** (B)
3. Venture capital: board seats allow monitoring of managers (principal–agent), and nonmarketable shares stop others from free-riding on the monitoring. VCs still pay to verify. (A)
4. Bondholders each hope others pay to monitor and enforce covenants (free riding), so too little monitoring happens; a bank with a private loan doesn't have this problem. Covenants don't enforce themselves. (D)
5. Manager perk: perk costs the firm $50,000, worth $12,000 to the manager. s = 12 ÷ 50 = **24%.** Below that, she gains from the perk. (B)
6. Collateral: safe = 1,200 − 1,100 = $100; risky = 0.5 × (2,000 − 1,100) = $450; C = (450 − 100) ÷ 0.5 = **$700**. Collateral is lost only half the time, so divide by 0.5. (D)
7. "Only people who don't need money can borrow" = adverse selection (lenders can't identify bad risks, so they favor high net worth or collateral). (B)
8. China: very high saving and labor moving from agriculture to industry drove growth; state-owned banks allocate capital poorly, limiting growth as the economy matures. (C)

Chapter 9

9. New deposit $1,000,000, reserve ratio 20%: deposits +$1,000,000, reserves +$200,000, loans +$800,000. (D)

10. Shortfall: reserves $50m, deposits $200m, outflow $50m, r = 10% → **$15 million.** Compute required reserves on the NEW deposit level. (B)
11. Order of fixes: borrow fed funds or sell securities first, call in loans last (antagonizes customers; loans are illiquid). A bank can't reduce deposits. (A)
12. ROA 1.2%, capital 8% of assets: EM = 1 ÷ 0.08 = 12.5, ROE = **15%.** (A)
13. Less capital raises ROE for a given ROA, and deposit insurance shifts losses away from owners, so banks hold too little capital and regulators impose requirements. (C)
14. Gap = 60 − 40 = +20; rates fall 3 points → **−$0.6 million**. (B)
15. Duration: assets $200m (dur 2), liabilities $180m (dur 4), rates +2 points → ΔNW = **+$6.4 million**. (C)
16. VaR = maximum loss over a period at a chosen probability; stress test = losses if an unusual combination of bad events occurs. Neither guarantees safety. Gap and duration are interest-rate risk, not trading risk. (A)
17. Compensating balance: acts as collateral and lets the bank watch checking activity (monitoring, reduces moral hazard). A credit-risk tool, not a source of high returns. (D)

Chapter 12

18. Financial crisis = a large disruption to information flows that sharply increases adverse selection and moral hazard, so lenders stop channeling funds to creditworthy borrowers. A stock drop alone isn't a crisis. (D)
19. Repo haircut 15%, loan $170m → collateral = 170 ÷ 0.85 = **$200 million.** (C)
20. Capital lost = $15m ÷ $24m = **62.5%**. (3.75% of assets is the wrong base.) (C)
21. Credit spread: peak 5.6, Y1 2.0 → increase **3.6 points**; wider spread = worse asymmetric information. (A)
22. Currency mismatch: net worth after = 55 − 5 × 12 = **−5 million pesos.** (B)
23. CDO tranches: pool loses $50m, equity $20m, mezzanine $40m → mezzanine loss = 30 ÷ 40 = **75%**. (C)
24. Microprudential vs macroprudential: if all banks sell at once, fire sales push down prices and every bank's capital and collateral, so the system becomes less stable. (A)

Chapter 13

25. Reserve Bank president is chosen by Class B and C directors (public), approved by the Board of Governors. Class A bankers don't vote. Not a political appointee. (B)
26. Fed independence would fall if Congress required its budget to come from annual appropriations. Testifying doesn't reduce independence. (A)

27. Rate cut from 5% to 3% before the election, then up to 6.5% afterward = **political business cycle** (the main argument for independence, showing inflationary bias). (B)
28. ECB mapping: Executive Board ↔ Board of Governors; Governing Council ↔ FOMC; national central banks ↔ Federal Reserve Banks. (C)
29. Bureaucratic behavior theory: the Fed is secretive and slow to raise rates when that would provoke conflict with Congress and the President. (D)
30. Bank of Canada has less instrument and goal independence than the Fed; many countries have granted central banks more independence. Don't confuse with the ECB, the most independent. (C)

Written Q1: managing a deposit outflow — reserves $26m, deposits $184m, r = 10%, $18m outflow.

- (a) Required = 10% × 184 = $18.4m; excess = 26 − 18.4 = **$7.6m**.
- (b) T-account: Reserves −$18m | Deposits −$18m. New reserves $8m; new deposits $166m; new required = $16.6m; **shortage = $8.6m**. (Not 18 − 7.6, because required reserves also fall by $1.8m.)
- (c) Sell securities: Reserves +$8.6m, Securities −$8.6m; liabilities unchanged.
- (d) With assets $200m, profit $2m, capital $16m: ROA = 1%; ROE = 2 ÷ 16 = 12.5%; EM = 200 ÷ 16 = 12.5. Owners prefer less capital because, for a given ROA, a smaller equity base raises ROE; the cost is a higher risk of failure.

Written Q2: boom and bust in housing

- (a) Looser lending standards → demand shifts right (D1 → D2), price rises P1 → P2.
- (b) Bought $300k, mortgage $270k. Peak = 1.20 × 300k = $360k; equity $90k. Bust: 0.70 × 360k = $252k; equity = **−$18k** (underwater). The bust is 30% of the peak, not of the original price.
- (c) Bubble bursts → demand shifts left, price falls to P3 < P2. Full credit for any clearly labeled leftward demand shift with P3 < P2.
- (d) Originate-to-distribute: the lender or broker earns fees and sells the loan in an MBS, so it bears little default risk and has a weak incentive to screen = principal–agent problem.

Part 7 — Formula sheet and trap checklist

Formula sheet

- Lemons price = lemon value (when good sellers' minimum > average).
- Average value = share good × value good + share lemon × value lemon.
- Manager perk: no gain if share × cost ⩾ benefit → share = benefit ÷ cost.
- Collateral C (lost only if the project fails, prob 0.5): Safe = payoff − repay; Risky = 0.5 × (payoff − repay); C = (Risky − Safe) ÷ 0.5.
- Fair insurance premium = probability × loss.
- New deposit D: reserves +rD, loans +(1−r)D, deposits +D.
- Outflow X: shortfall = r(D − X) − (R − X). Excess falls by X(1−r). Max outflow with no changes = (R − rD) ÷ (1 − r).
- Cost of excess reserves = excess × (loan rate − reserve rate).
- ROA = profit ÷ assets; ROE = profit ÷ capital; EM = assets ÷ capital = 1 ÷ capital ratio; ROE = ROA × EM.
- Gap = RSA − RSL; Δprofit = gap × Δi (decimal).
- ΔA = −dur_A × Δi × A; ΔL = −dur_L × Δi × L; ΔNW = ΔA − ΔL.
- Home equity = value − mortgage; % change = Δequity ÷ original equity; bust % is of the peak price.
- % of bank capital lost = loss ÷ old capital.
- Real debt = nominal ÷ new price level.
- Repo: collateral = loan ÷ (1 − h); max assets = capital ÷ h.
- Foreign-currency debt in home currency = foreign debt × exchange rate.
- Credit spread = Baa − Treasury; report the increase. TED = 3-mo interbank − 3-mo T-bill.
- CDO: subtract equity first, then divide by the mezzanine size; senior is hit last.

Trap checklist

- Largest single U.S. funding bar = nonbank loans (38%), not bank loans.
- $9,000 is only the first offer. Lemons price falls to the lemon value.
- Collateral formula: divide the payoff gap by 0.5.
- Smaller manager stake = worse principal–agent problem.
- Compute required reserves on the NEW deposit level; excess falls by X(1−r), not X.

- A bank can't "reduce deposits" to cure a shortfall. Calling in loans is the worst option.
- Selling securities changes only the asset side; the bank swaps one asset for another.
- Equity multiplier = 1 ÷ capital ratio (not the ratio itself). Higher capital lowers ROE.
- Gap and duration measure interest-rate risk, not trading risk. Positive gap: profits fall when rates fall.
- Rates in formulas are decimals (3 points = 0.03).
- Longer-duration side loses more when rates rise; if liabilities are longer, net worth rises.
- Compensating balances are a credit-risk tool, not a high-return tool.
- Crisis = breakdown of lending, not just a stock decline.
- Haircut: collateral = loan ÷ (1 − h). Don't add h to the loan.
- Capital lost is measured against old capital, not assets.
- Credit spread: give the increase, not the peak level.
- Housing bust % is of the peak price, not the original price.
- Tranches absorb losses in order (equity → mezzanine → senior), not evenly.
- Microprudential (one bank) vs macroprudential (whole system); fire sales are the blind spot.
- Chair has no veto. 5 presidents vote (NY + 4 rotating), not all 12. Fed budget doesn't depend on Congress; annual appropriations would reduce independence. Testifying ≠ losing independence. Member banks don't control the Fed.
- Bank of Canada has less independence than the Fed; ECB is the most independent and its goal is in a treaty.
- Classification shortcuts: can't tell risky applicants = adverse selection; secret switch to a riskier project = moral hazard; copying others' research = free rider; insurance requirement in a loan = restrictive covenant.